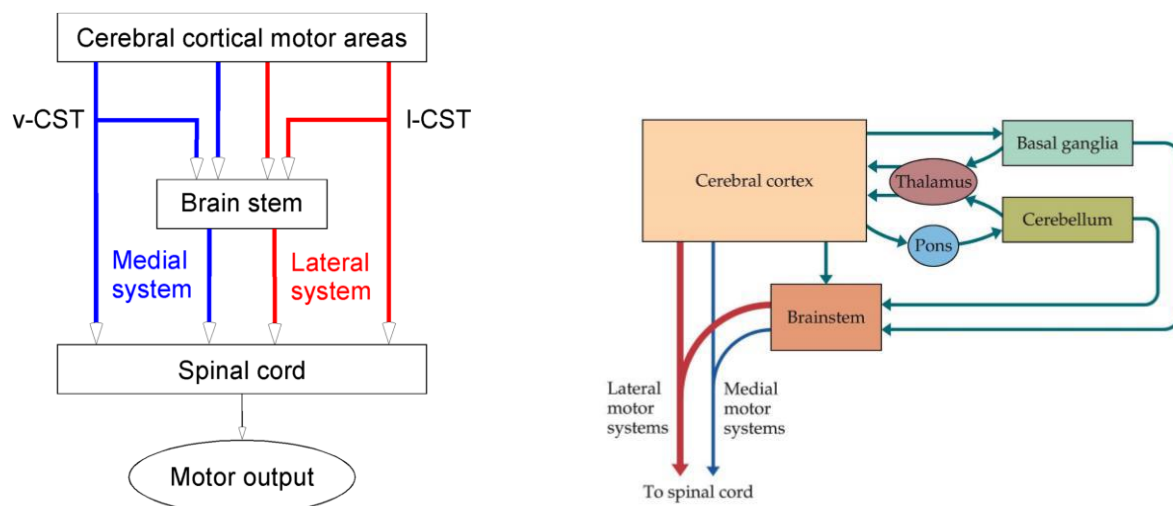


and define a taxonomy or features that represent and characterize action, we fail. This is because when we consider action per se there are plenty of features to be considered, and most of the time when we consider action as a whole we have other cognitive functions that are involved (if I have to grasp something I first have to perceive the object and understand the features and how part of my body, what we define an effector, is actually interacting with the object, and even the way we are going to interact with it). One other message that is a sort of feedback message we got from this, is that all different aspects of our cognition (including perception) are influenced by the motor activity that can be acted upon a specific object.

If we focus on the effector, the way our body interacts with an object depends also in the way I represent it in my body, and at the same time the possibility of actions that my body could act upon the specific object depends on its position. So, it's really difficult to comprehend a classification of actions, based on the different levels to which you can describe it (action can be transitive or intransitive). This is something that we can describe semantically, but if we go on a clinical perspective, you find a different classification, and this is really getting different levels of taxonomy that make hard to understand how we really process with action and also tells you how why AI based systems fail classifying action. It's very difficult to deal with an unpredictable event.

We are going to go briefly to describe different levels of representation and just see a flavoured glimpse of how we can start from motor commands going to action representation and processing according to different levels of representation and primarily figuring out the whole brain is involved in action processing.

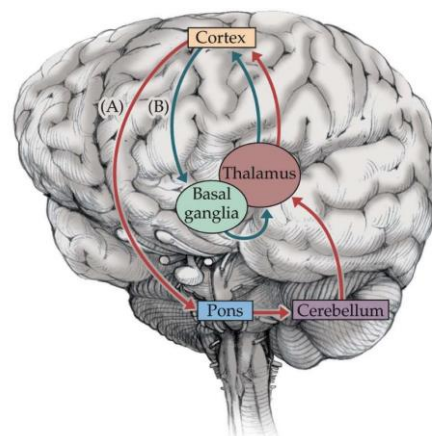
When we talk about the general organization of the motor system, we could obviously get different levels of representation. Our first aim is to have some specific piece of information that originates from the motor cortex and has to be forwarded to the peripheral of the body, and primarily to muscles that allow us to enact specific behaviour. This process is offered very effortful and time consuming, and indeed we have also motor components that are enacted independently of the commands from the central nervous system, there are specific behaviour that for the sake of survival are enacted without being conscious of that at the level of the spinal cord (central pattern generators) that regulates the simple movements. When there is a damage in this pathway, the command cannot be fostered from the cortex down to the muscle, but if there are connection there are still valid at the level of the spinal cord, for relatively simple commands (like locomotion), you can stimulate spinal components to induce very simple motor elements, and the sequential stimulation can allow to put together different movements and components into a simple finalized act.



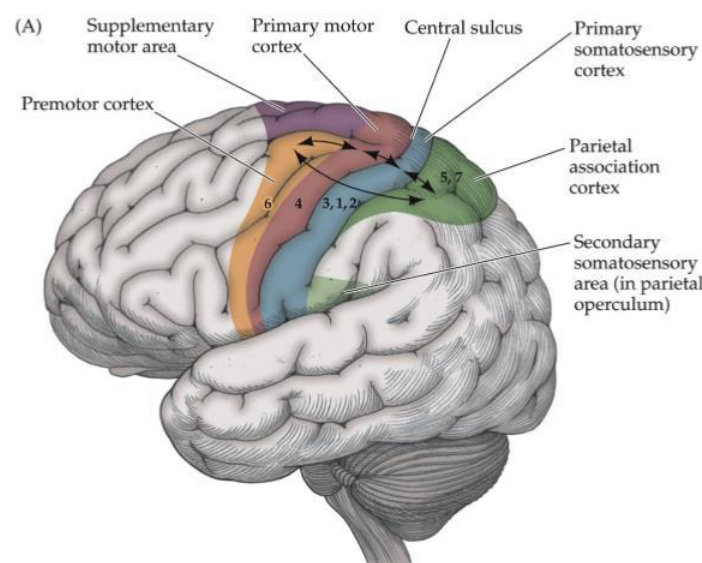
The most difficult part of our motor system, is that we have not just to operate as a sort of like simple element, but for instance in order to still fruit towards something I have also to use other muscles to stand up, so there should be some processes that should be come somehow automatic, and there are plenty of movements that we are able to perform that are not actually under voluntarily control, but are performed without really conscious processing or representation of those, and the more you become skilled in a specific movement the more effortless the movement becomes.

One thing clear is that in this chain of acts controlling motor functioning, there are two things that we have to highlight:

- There are different levels, and this is hierarchical (cortical motor areas -> brain stem -> spinal cord, directly having specific motoneurons having an acetylcholine synapsis with the muscle).
- At all these stages there are specific feed forward and feedback processes.



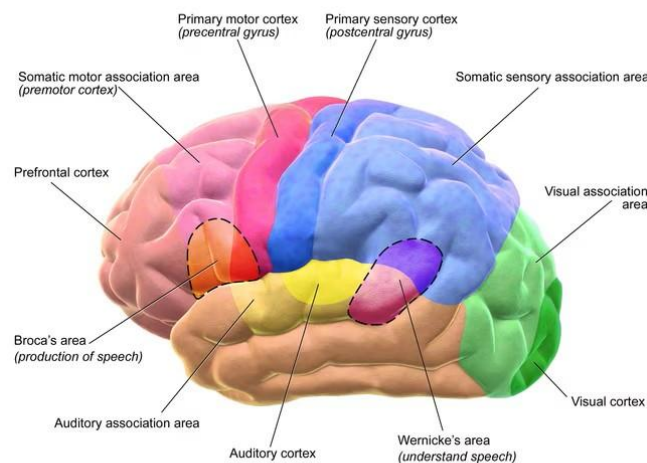
These refined movements are actually under the control of different acts, that have constantly this loop of control, that primarily guarantees that when you decide to perform a movement it's fluid and is constantly updating. The most skilled person are athletes that are able to predict in advanced some specific behaviours.



The primary motor cortex is where we have these pyramidal neurons, also the motoneuron α , that is the biggest one in the cortex with its long axon, and they project downwards primarily to the spinal cord. But some of these neurons are actually going and arriving just at the brain stem, so if

we look at the cortical level, most of descending pathways that start from the cortex get from the Brodmann area 4 (the pre-central gyrus), but this is not the only region that is able to send projection down to the spinal cord, because 80% come from the Brodmann area four, but the other 20% also come from other regions. More anteriorly we have the pre-frontal cortex, and anterior to the pre-central gyrus we have the pre-motor cortex, primarily divided into a dorsal and a ventral portion, and the ventral portion is almost w shaped and this where we have the inferior frontal gyrus, that has this m or w shape, and here there is the Broca area of language but also the ventral pre-motor cortex.

Motor and Sensory Regions of the Cerebral Cortex



More immediately we have the supplementary motor area, that appears to be fundamental to process bimanual coordinated processing. The pre-motor cortex (both ventral and dorsal component) appears to be much more fundamental for processing the goal of the action.

The precentral gyrus is also mirroring the postcentral gyrus, that is where the primary somatosensory cortex is representing different body parts, location and mechanical stimuli getting to the skin. This piece of information is projecting posteriorly to the Brodmann area 5 and 7 of the parietal cortex, where we have several portions of the cortex that are very interesting in order to represent action, but here we got information from the motor component with the visual component about space, so in the parietal cortex we'll have actions primarily representing according to their location in space.

The eccentricity bias is the idea that the retinotopic organization is projecting beyond the early visual area not just ventrally but also dorsally, so clearly the way I'm representing actions depends on the way it's located in my visual field, and there is this tight coupling between the motor representation of an action with what is typically the representation and location in the visual field.

If some of these regions appear to be sending very simple commands about motor planning, these other pre-frontal and parietal regions appear to be much more related to a sort of higher order representation of motor planning.

In the portion of the dorsal and the superior frontal gyrus there is another area called the frontal eyes field that primarily regulates the movement of the eyes, and this region in the frontal portion is actually related in moving the eyes accordingly to perform some specific actions.

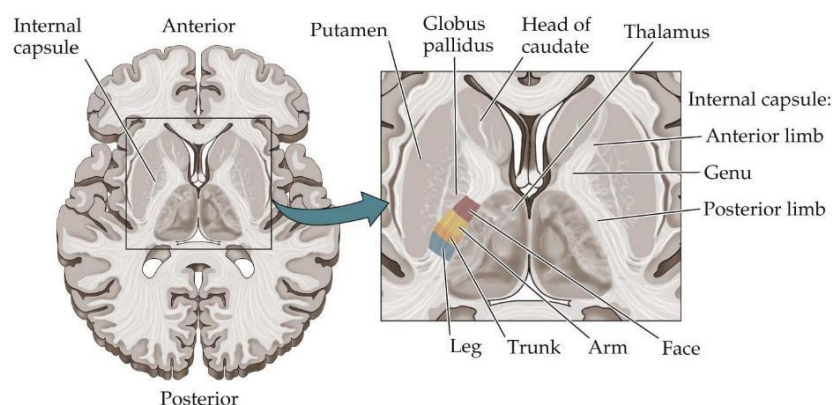
Here in the superior parietal cortex there are single patches dealing with different elements, so if I have a disorder in a certain point I have a disorder called agnosia (inability to recognize object identity), and when I'm going dorsally I may have two kind of lesions; one that is called ataxia but much more optic ataxia (I'm able to recognize the object the object and decide which action to perform upon it, but I'm unable to perform a precise reaching space).

One of the features that is very similar between the somatosensory cortex and the motor cortex, is the representation as the homunculus, even with some differences. The motor component is much more enhancing the effectors, the representations of feet and fingers are much broader in the motor homunculus as compared to the somatosensory one, so it's mirroring physically some aspects of our body, how we are effectively able to interact with the surrounding world. This kind of somatotopic organization at the beginning has been described by a neurosurgeon (most famous one Penfield), putting electrodes and trying mirroring the somatotopic organization, but this is mostly true for the most posterior portion of the motor cortex, because the more anterior one is actually already trying to process action much less as organized as from a somatotopic perspective but much more relative to a sort of aim of the action.

The information from the cortex is in a constant feed forward and feed back loop with the thalamus and the basal ganglia, together with the brain stem, that is typically then forwarding information at the level of the pons of the cerebellum, that typically receives information at the level of the cortex and it goes to the nuclei of the cerebellum, and finally the information is projected back both to the brain stem and then to the basal ganglia. This loop of motor control highlights two aspects:

- Motor commands are redundant, we have to be confident that if we have a specific goal, I am able to perform that.
- There are several mechanisms that control that the whole process is controlled smoothly and uneventfully.

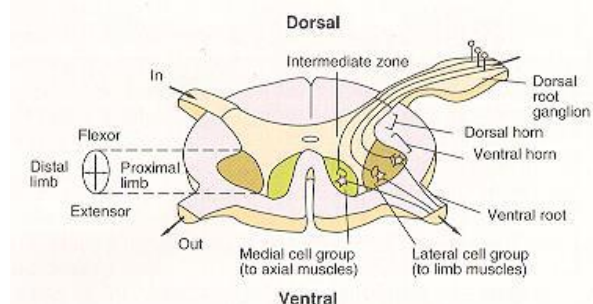
The first kind of control is in the basal ganglia, where we have several structures, like the caudate and the putamen (together they form the striatum, that releases dopamine, and disorders at the level of this substance are causing Parkinson syndrome), the thalamus and the globus pallidus. It's important to know is that all fibres that from the cortex are going down to the spinal cord, are actually passing in what is called the internal capsule, that is separating the caudate from the putamen and is also separating the pallidus from the thalamus, and we can also separate what is called the anterior limb, the genu and then the posterior limb. According to the organization of the fibres we have a specific representation that is maintained also somatotopically, so there is a topography also of the fibres that from the cortex progresses downstreet to the spinal cord (we have the leg, the trunk, arm and face).



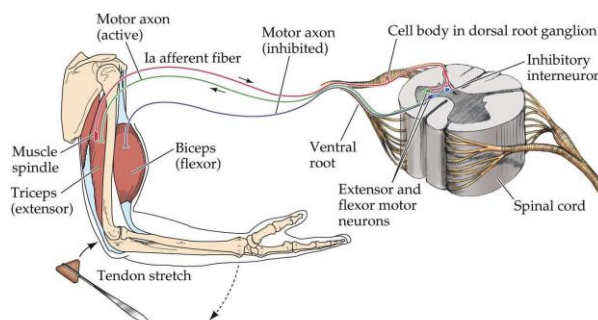
The information from the cortex to the internal capsule is then getting down along the spinal cord and we have also fibres that progress from the spinal ganglia down to the spinal cord. We are going to separate two systems, the lateral motor system and the medial motor system. The first one is primarily fed from the cerebral cortex, but there are also fibres from the basal ganglia and the brain stem, and this system is also called the pyramidal motor system and is primarily related to the regulation of voluntary movements. The medial motor system is also primarily fed by the basal ganglia and the brain stem and is primarily related to regulating all the muscle chain of the median part of the body, the central part of the body; these are movements that are not voluntary and are primarily dealing with regulating automatic movements, cyclic movements, and maintain posture and balance. Any disorder that affects this pathway (called extrapyramidal or medial pathway), either because of a lesion or of the source, impacts posture, gait, so the fluidity of movements, even voluntary ones.

At the beginning they were primarily described as pyramidal or extra-pyramidal, pyramidal because pyramidal neurons are the motoneurons that are present in the primary motor cortex, but since the cortex is contributed to both pathways, we rather describe them as lateral and medial motor system, that mirror their position in the spinal cord.

The dorsal horn of the spinal cord is primarily sensitive, with information getting from the periphery to be forwarded to the post-central cortex as part of the somatosensory cortex, and the anterior portion of the spinal cord (the grey matter of the spinal cord) is instead related to motor, and the medial pathway (in yellow) is primarily related to the axial muscles, while the lateral portion (in orange) is primarily sending information to the limbs, the effector, to the periphery.



Here typically we have a motoneuron that gets information from the spinal cord to the ventral roots, to the muscles. Here we have a very simple stretching reflex loop, because information at the level of the spinal cord is not just afferent to muscles, but the muscles spindle is sending projections and information to the spinal cord about the position of the muscle and the posterior sensitive elements is connected to the anterior one, through the inhibitory interneuron that connects the two neurons. So, the ventral nerve roots convey efferent motor signals, and the dorsal roots are getting the afferent sensitive signal.



Typically, the muscles spindles (sensory receptors that detect the amount and weight of stretch in muscles) transmit information to the distal processing of sensory neurons and convey to the spinal grey matter through the dorsal root. Here we have the inhibitory interneuron that connects to the anterior motoneuron, and this project to the muscle. So, this local circuit controls what you typically experience, if you want to extend your arm you have to use one group of muscle, but you should inhibit the antagonist group of muscle that is contrasting the action. When the neurologist hit the tendon on the knee, it stretches the spindle and you are performing a stretch and providing information that that part of the muscle is stretching, and this is going to induce an opposite reaction (the extension of the leg). So, what you are actually testing is this type of circuit, that has to inhibit or excite agonist and antagonist muscles of our limbs. These kind of stretching reflexes, are also useful when the stimulus is not starting from the spindle but is starting from the skin, just as with pain, where you have a painful stimulation, and before the information gets to the cortex you just remove the limb from the possible source of damage.

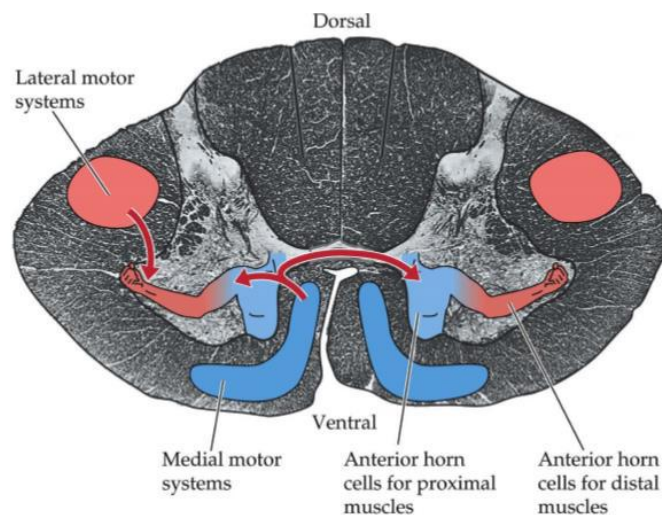
The control effect that the cortical motoneurons are performing to the spinal cord is very important, because is regulating the excitatory inhibitory balance. Most of the time, when you have a lesion at the level of the spinal cord, and this is no more able to get information from the brain, muscles are going in a sort of like conflicting presence of stimuli of excitatory and inhibition, and without this element of control you see that the muscle is primarily stretching and detecting the position that we call plasticity, is excessing excitatory response.

This kind of complete pathway from the cortex down to the spinal cord is fundamental to maintain the proper functions of the muscles.

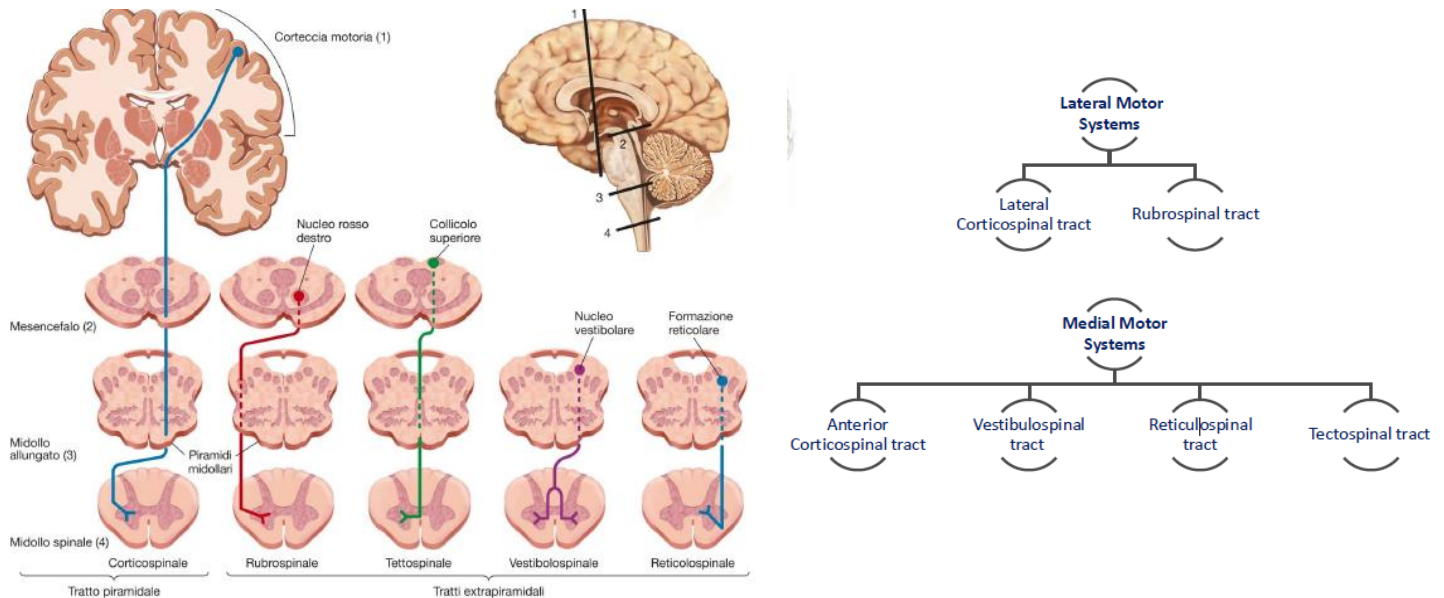
Motor pathways

At different levels the lateral and the medial pathways are not just maintaining the somatotopic organization, but at the same time they cross the spinal cord, and they reach the opposite body. So, similarly to what we have seen describing the tactile component, the right hemisphere is controlling the left part of the body, and the left part of the cortex is controlling the right part of the body.

The majority of the fibres of the cortex (85%) is creating what we have been calling the lateral pyramidal projecting system, but the 15% of the cortical fibres are also going to form the medial extra-pyramidal tract, and this is typically how we do not just enhance the somatotopic organization per se, the more anterior component pathway is processing the medial part of the body, while the lateral motor system is precisely representing the limbs.



We have the lateral motor system and medial motor system, the first one is made up by the cortical spinal tract and what is called the rubrospinal tract (originating in the brain stem, in the mesencephalon, and is called the red nucleus), that immediately crosses fibres and goes down. Then we have other four tracts, that are actually making up the medial motor system. The cortical spinal tract (15% percent of fibres originating from the cortex), the vestibular spinal tract, the reticulospinal and tectospinal. These are all brain stem nuclei that project down to the spinal cord. See the slides to see better each one of them.

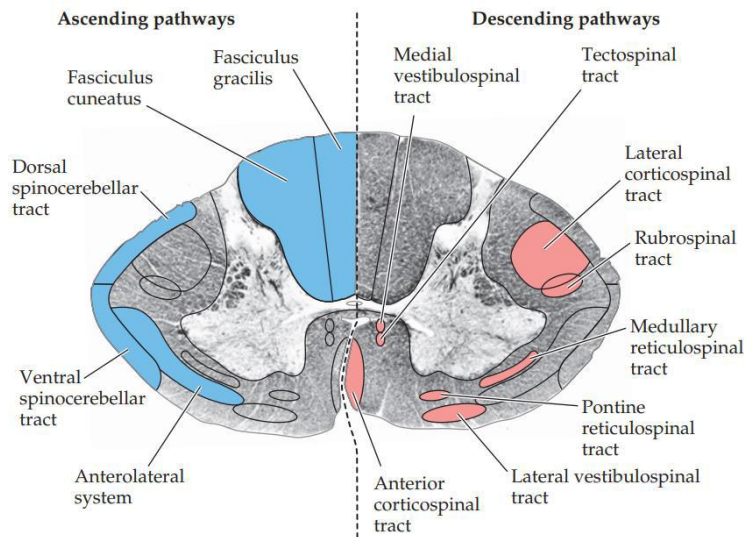


	Site of origin	Site of decussation	Levels of termination	Function
LATERAL MOTOR SYSTEMS				
Lateral corticospinal tract	Primary motor cortex (precentral gyrus) and other frontal and parietal areas	Pyramidal decussation, at the cervicomedullary junction	Entire cord (mainly at cervical and lumbosacral enlargements)	Movement of contralateral limbs
Rubrospinal tract	Red nucleus, magnocellular division	Ventral tegmental decussation, in the midbrain	Cervical cord	Movement of contralateral limbs (function uncertain in humans)
MEDIAL MOTOR SYSTEMS				
Anterior corticospinal tract	Primary motor cortex and supplementary motor area		Cervical and upper thoracic cord	Control of bilateral axial and girdle muscles
Vestibulospinal tract	Medial VST: medial and inferior vestibular nuclei Lateral VST: lateral vestibular nucleus		Medial VST: cervical and upper thoracic cord Lateral VST: entire cord	Medial VST: positioning of head and neck Lateral VST: balance
Reticulospinal tract	Pontine and medullary reticular formation		Entire cord	Automatic posture and gait-related movements
Tectospinal tract	Superior colliculus	Dorsal tegmental decussation, in the midbrain	Cervical cord	Coordination of head and eye movement (uncertain in humans)

The reticular system is primarily involved in regulating arousal, and the superior colliculus is receiving information from the retina but is also receiving information from others sensory modality and is fundamental as a first hub of multisensory integration. More importantly, is also sending commands to very simple elements of motor effector, but this explains why coordinates end in the eye movement, because the superior colliculus is primarily related to perception and

integrating perception from different modalities. So, this small nucleus in the brain stem is coupling perception with movement.

So, the main subdivision is between a system that appears to be regulating the most peripheral part of the body and voluntary movements, and another system (the medial one) much more regulating sensory motor coordination, gait, posture and balance.



Action is really made up of several parameters and if we have to look at different action features, and neuroscientists have always been describing action as a sort of hierarchical processing of information and commands, we can describe the kinematics, the effector, the effector target interactions, the target features, the call of the action, the outcome and several others. So, there are several pieces of information that we could try to represent and process at the action level.

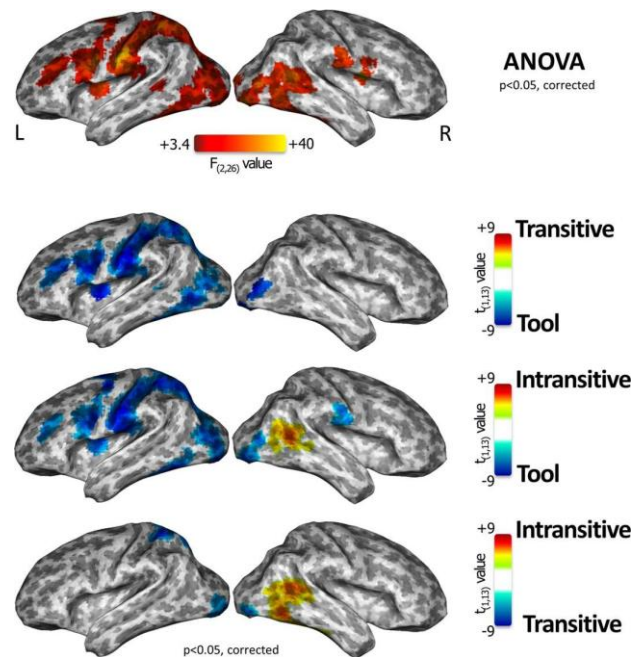
If you look at all the different kind of experiments, this definition is really telling us that is really difficult to have right comparisons in the tools that are available in neuroimaging, trying to disentangle regions specific for trajectory, for a goal, an object, because these regions are actually overlapping, so one thing that you should imagine is that these regions appear to represent all piece of information relative to action.

They tried to use recognition of three different kind of actions, tool-mediated, intransitive (action more social) and transitive (like grasping). If you look at which action is represented, you find that motor cortex most of the time is hardly representing, so when we are perceiving action we are not relying on motor cortex per se, but there is a wide gamut of brain areas that appear to be involved, and several of those are actually involved in representing actions and are the posterior parietal cortex, the supplementary motor area, dorsal and ventral premotor cortex, and there are also portion of this superior temporal vision cortex.

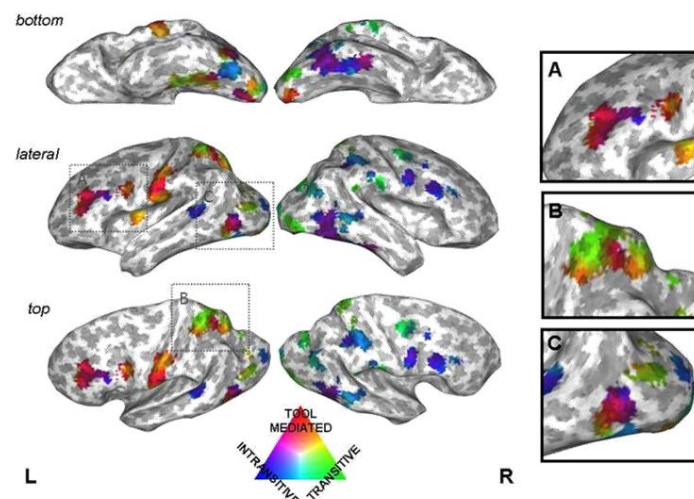
If we look at the comparisons of two mediated vs. transitive, etc., clearly, we are unable to find which regions are specific for each class of action, the whole brain is processing information.

You can see a sort of map that in a better way represents the different representations of action classes, and the same idea that we have been presenting about the topography of object form (pattern of response for different categories) was done here on action representation. This kind of separation allows to capture specific elements that are of interest; first of all, when you have a

representation of intransitive action and you have to represent objects, you see that the ventral pathway is deeply involved, because obviously we need to have information about objects here. The same thing happens for tool-mediated actions.



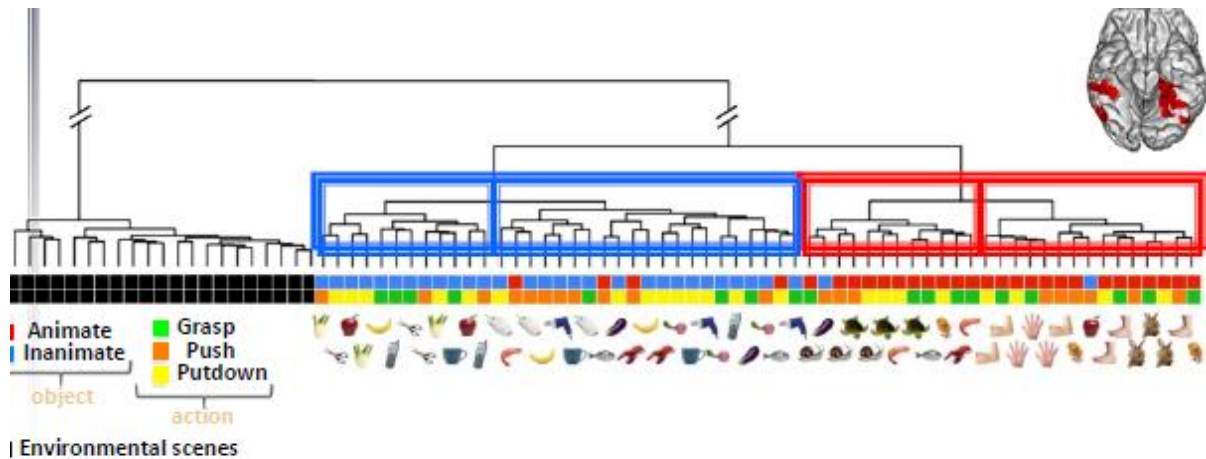
In the p-pattern in the superior parietal cortex we recognize a sort of gradient, that mirrors the representation the more transitive in the periphery and the tool-mediated, so there is some kind of idea of how we can represent action differentially. And the left premotor cortex, the left inferior parietal cortex, is actually processing here tool-mediated action.



So, for sure this kind of representation of the action, and here we are just performing the perceptual component of that, is telling you that gathering together different features involved in action representation, we can separate different classes of motor behaviours, and this is just telling us that action representation, as a distributed pattern of the cortex, is actually representing all these features into a single unique representation, as we were observing for the ventral stream in visual cortex. The brain is actually integrating different piece of information into a unique representation distributed on a broader network of action representation, something that we can call as action representation network, that involves frontal, parietal and occipital temporal area, and this broad network appears to allow the brain to separate and distinguish interaction classes

and then each specific action. For example, the ventral temporal cortex will process the identity of the object on which we are performing the action.

In a study three actions were performed, and each action had to be performed either on an animate (in red) or inanimate (in blue) object.



- Actions vs. environmental scenes (Accuracy 90.1%, $p < 0.0025$, chance level 50%)
- Animate vs. inanimate objects (Accuracy 68.3%, $p < 0.0025$, chance level 50%)
- Bodyparts vs. animals vs. vegetables vs. tools (Accuracy 37.7%, $p < 0.0025$, chance level 25%)
- No classification of motor acts (Accuracy 35.2%, $p < 0.8$, chance level 33.3%)

The first kind of classification tried to have was to separate action stimuli (a hand performing the three actions) vs environmental scene, so the classifier has a huge accuracy. Looking at a short stimulus, how brain is able to separate action vs. environmental stimulus.

One thing that was clear, was that the ventral temporal cortex recognizes objects, so we can recognize animate vs. inanimate. Objects were identified irrespective of the actions they were acted upon. So, the ventral stream is focused only on recognizing the object, the goal of the action. Also, one thing interesting is that there is no classification of motor acts from the ventral stream.

The parietal cortex is actually able to classify action vs. environmental scene and is also able to classify the different action classes, but it separates the grasping but is not able to separate the push from the putdown. The parietal cortex acts and relies on a special component, grasping is something that approach the object to me, the other two are putting the object away from me, so there is a spatial component that changes. The parietal cortex is also able to separate animate from inanimate and this is because information from the ventral cortex about identity are also conveyed to the dorsal pathway. So, while the ventral stream is actually able to discriminate several pieces of information about the object classes but no motor component, the parietal cortex starts processing and separating some action classes, maintains some high level discrimination, but is losing the detail of information that the ventral temporal cortex has, it doesn't discriminate body parts from animals or vegetables, the information in the parietal cortex is primarily related to the motor action represented in space.

The prefrontal cortex is doing something similar, but even more detailed. The pre motor cortex should really get represented the goal of the action, and is able to separate with detail all the

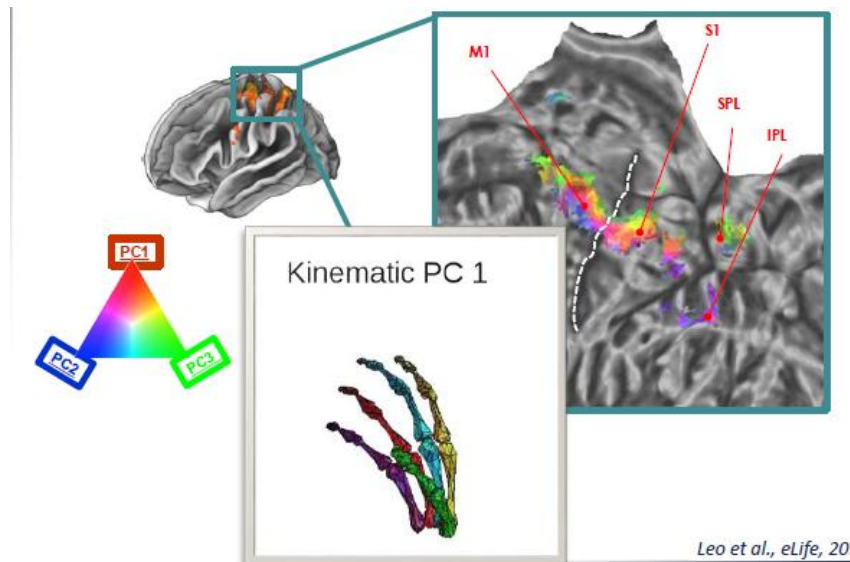
comparisons, so is much more detailed about the identity of the object to which the action is acted upon, but at the same time is able to classify much better grasping from putting down or pushing, so we have more information about the details of action. We are also able to discriminate whether is grasping an inanimate or animate object, so the premotor cortex is getting together the information relative to the identity of the object and the aim of the action, because representation is getting more meaningful in the premotor cortex.

The different portions of this circuit appear to subserve different pieces of information of action processing. So, the motor representation of an action appears to be more distributed broadly at the level of the cortex and specifically depicting information of the action as a whole. This actually relates to something we were saying before, action is processed as a whole and the different components also have unique representation according to action class and identity, and this is also reconciling with the idea that the whole brain is working to represent the different actions.

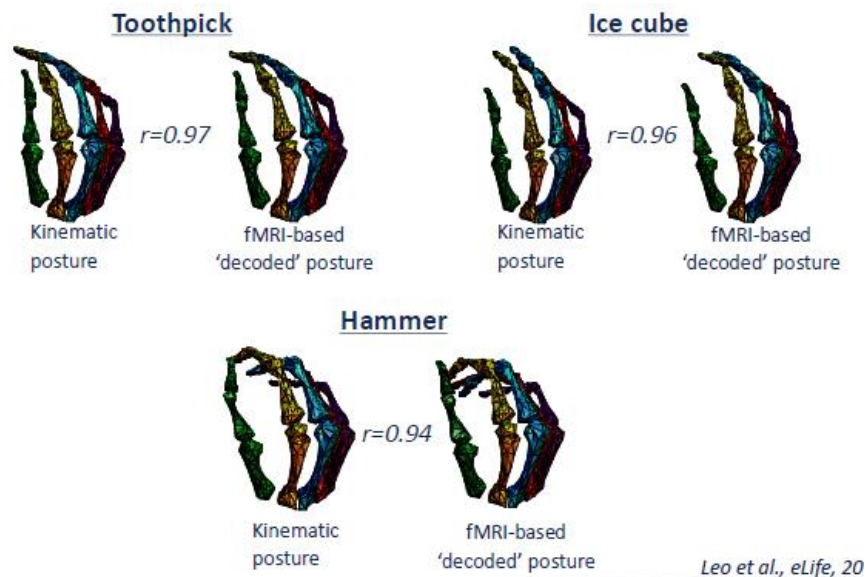
But how we really jump from grasping an object to someone else doing it?

One element that we have to consider is how much important is the kinematic and the motor information, so how does the human brain control movements to carry out different tasks. The message from the cortex to the effector is redundant and has a huge cost transmitting the information since our effector is a very complex tool. The central nervous system manages and controls movement of individuals muscles and joints with 23 degrees of freedom. Synergies represent functional modules that includes cells of hand muscles and joints which are recruited simultaneously and act as a unit. Typically, the idea of reducing these 23 degrees of freedom to a lower number of dimensions is telling that probably every kind of grasping could be reduced to a combination of specific joint angular movements and muscles recruitment, and this is also something that typically has been suspected with the first neurosurgeon dealing with the electrostimulation of the cortex. You may have to try and move you little finger without moving the ring finger (difficult), obviously if you pay attention and you exercise you can adopt some strategies, but nonetheless there is some kind of synergy and this because they are the less used fingers in order to perform any kind of action. The idea of synergy is reducing a lot the number of information, and the combination of different synergies can adapt to any type of grasping movement. When you decide to reduce from the principal components analysis, only the first 4-5 components allow you to describe about 95% of all your final hand postures.

It was tried to record 20 different gestures with optical tracking of muscular and neural components. The images don't tell us which brain area actually activates, but in which brain area this kinematic information is actually represented. You are doing what is called encoding, you take the matrix that you create according to your kinematic data and define hand posture, and you use this to encode whether during grasping these components are represented at the brain level. What you find is that these regions are all actually involved during the performance and you see that the kinematic component is present in the precentral cortex, in the postcentral cortex, in the posterior parietal, in the supplementary motor area and in the dorsal and ventral premotor cortex. All the regions are representing these kinematic loadings and if you try to map this you see that the three component that are reproducing 80-85% of the brain are actually reflecting; the different components are moving the five fingers together, they are nicely and topographically represented at the brain level. This explain how in the hand area we have a very nice representation at the level of synergistic movement have that appears to be topographically represented in the cortex. It's interesting because the motor component appears to be present at the level of the representation.



You can also try to use brain activity to reconstruct specific postures. If you are able to recreate a specific gesture from brain activity you can convey this information to any kind of external device.

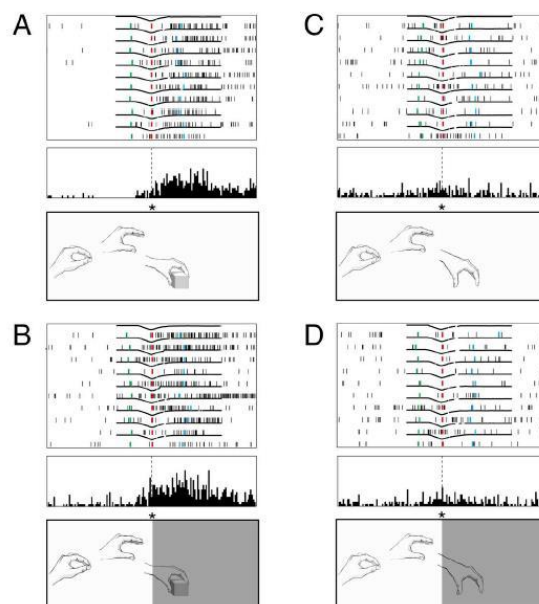


So, if kinematic synergies appear to be topographically organized within motor cortical area, what about perception? They tried to reproduce the same kind of experiment also during motor imagery, so subjects weren't performing any actions. You can see similarity in the pattern primarily in the premotor and in the posterior parietal cortex. This is telling us that there is a common language, a motor information, that share between motor execution, performance, planning and recognition.

There is motor information that links and associate motor performance to perception, and this is fundamental because all this organization is broadly impacting our representation of the external world and the interpretation of other people behaviour. There is a strong demonstration that motor information is mirrored in performance and representation of an action.

Mirror neurons (I don't have the first part)

If I would record eye movement, as soon as I'm going to pre-shape properly, your eye movement would direct with the touch from my head much in advance before I'm reaching the object and would go for either the bottle or the pencil. The same did the monkey, you show a cube, you show the hand resting, pre shaping and grasping, with the monkey that recognized the movement preparation and grasping. Is that related to object per se? No, the same thing could be done like in another experiment, the monkey is showed a cube, then the cube is obscured, the monkey sees the hand pre shaping and then is not actually seeing the last grasping, but the monkey know that the cube is there and it sees the pre shape, so that is going to grasp, the mirror neurons discharge. So, here is not related to the presence of the object or the kinematics. You can perform the same thing but without the object, and this is telling you that the action is not a finalized one, so the mirror neuron is not discharging, as for the control condition.



Therefore, mirror neurons appear to represent the aim of a finalized action, the meaning of the action, but not its specificity. Together with mirror neurons, the researchers were studying another family of neurons, canonical neurons, that are slightly different from the first ones, because, in the same region where mirror neurons are, canonical neurons appear to be activated when the monkey motorically grasps the object, or when the monkey sees the object without any action. These neurons are something that we have been describing in the dorsal pathway, they are neurons that respond to the object and when I have to specifically grasp the object.

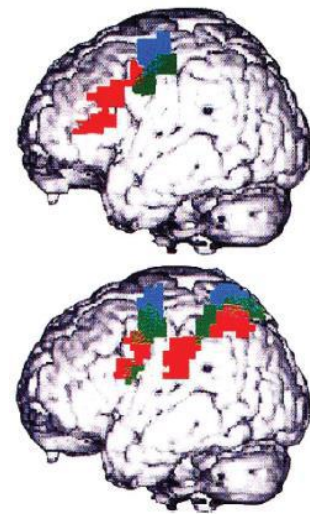
Mirror neurons appear to be activated when you perform an action and you see another agent performing the same action, canonical neurons appear when you perform an action on an object or when you see the object without any action, So, canonical neurons appear to be more primitive, simpler as a conceptual representation, mirror neurons appear to be conceptually little higher in the hierarchy of representation, they represent the finalized action per se. But, how we really 100% confident that it doesn't relate with the kinematic? Because kinematic matter is really widespread in that portion of the cortex.

In another research pliers were utilized, where in order to grasp you have to close the fingers to press an to use this tool, but there are also reverse pliers, and here in order to grasp you have to open your hand. Kinematically these two actions are opposite. The same neurons that are telling you to grasp the plier is responding only when you grasp the plier independently of the kinematic

of closing or opening. So, mirror neurons appear to be totally unrelated from the kinematic motor aspect, appear to be a different way to processing it. When the monkey observes someone else opening the peanut and grasping the fruit, when you just see without even hearing the cracking of the peanut, but since this representation is independent of the kinematic aspect you also see that the monkey is not actually seeing the movement of the peanut breaking but it's hearing only the noise. So, specifically this representation is so abstract that even the sound is actually working to represent the action. For example, if you hear a ring your brain is also processing how to ring the bell.

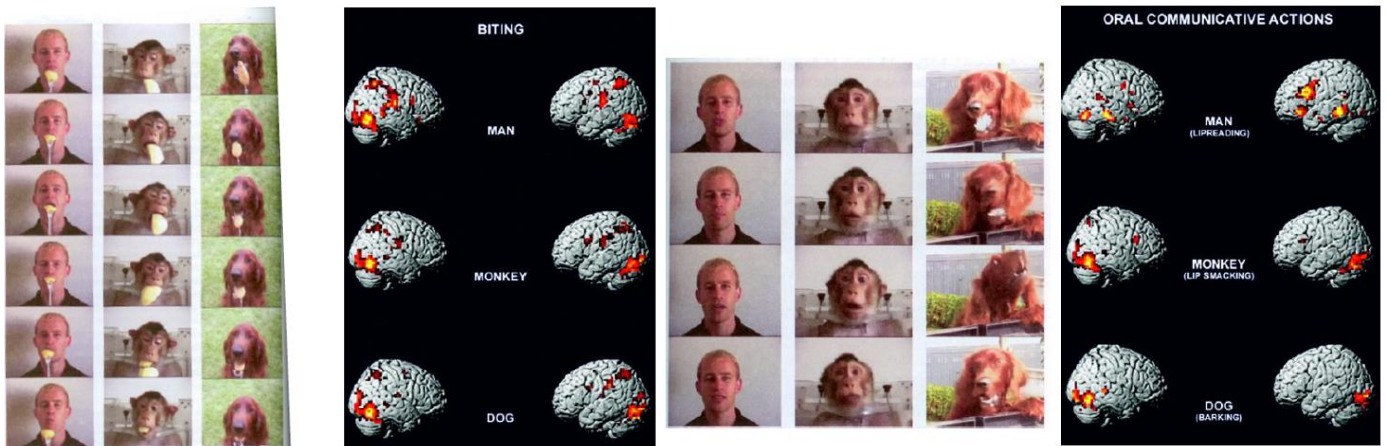
When describing mirror neurons, we were finding mirror neurons that were called as broadly or narrowly congruent. There were mirror neurons that were responding to any kind of grasping (broad mirror neurons), or very specific grasping. This should remind us of the representation in the premotor that we were describing before, the ability to classify a label of grasping animate or inanimate objects.

Mirror neurons have been extensively described in monkeys, but we soon got to humans. Here the goal was to describe what is called the mirror system, and it is also interesting because the mirror system in humans is also somatotopically organized, here we have mirror neurons response of the feet, of the hand (in green), and of the mouth and lips (in red). These mirror neurons are really closed to the inferior frontal gyrus, where the broca area is. Mirror neurons in humans have a much broader network then in monkeys, in humans we are able to code not only transitive actions but also intransitive, because I have language development. Typically, it codes the sequence of movement of an action, so most of the time these actions in the mirror neurons are very complex and, off note, you can also activate mirror neurons when you mimic an action you are familiar with.



Mirror neurons are also important because we learn through imitation, and it's the first way we actually learn motorically, and our brain is tuned and tailored. We tend to imitate and recognize other people, but the idea of imitation is also what allows us to enter into other people shoes and understand their intentions, and this is not just motorically, is also creating what is called empathic, the idea of sharing feelings with other people. This idea is important because it allows people to share common sensory motor repertoire.

In a study there were shown 3 movies, and while viewing them people were recruiting bilaterally parietal and premotor regions, the same action of biting food is shared between man, monkey and dog. We can see without listening a man talking and read lips, the monkey opens and closes its mouth, and the dog bark. While doing these actions, only while viewing people talking, we were recruiting premotor mirror neurons, but not for the dog barking, because barking is not in our motor repertoire and this is what is clear with this idea of having something experienced sensory motorically. In order to understand other people intentions you need to have experienced that.



In a comparison between capoeira dancers and classical ballet dancers both were danced while presented both some video, but mirror neurons were responding higher when each of them were shown videos of people dancing their style.

This is very important in order to actually understand how motor information is really rooted in my perception, and how it allows me to predict in advance, and also the least I know some motor condition the less I'm able to predict that specific behaviour. When you see and are familiar with an action, while I'm doing that and someone is watching if I would record their muscle, they would be ready to enact the movement, and this is also a strategy used for therapy, exposing you to movements that you are not able to perform anymore, in order to have your motor system to get ready to represent and retrieve that kind of motor action. Also, the context of an action can tell us how we can modify.

Mirror neurons too are present in blind people, so people that have never seen are able to have mirror neurons system responding to the recognition of action, and even their mirror neurons system was responding more to action they were familiar with.

Mirror neurons appear to be a sort of like of empathic too; if I'm able to enter other people shoes and understand their intentions, I'm also able to understand their feelings and when you look at regions that are involved in empathic you see that is plenty of regions that are present in the mirror neurons system.

Anthropomorphism is a feature you can play with in your device; in order to build something that your brain is able to embody in your body schema, because we recognize it most.

These aspects will be fundamental if you are going to look at this in terms of embodied cognition; the idea that every time you have to perform an action, every time we have to process a stimulus or try to have a broader conceptual representation, this is rooted somehow in a sensory motor experience. How can I work with a tool or device to be embodied with the patient, to be truly part of his body representation, part of his motor repertoire.

Plasticity

Plasticity has been defined as an intrinsic property of the human brain, representing a sort of invention of evolution that enables the nervous system to overcome the constraints imposed by its genome, allowing it to adapt to the environment or physiological changes and experience.